



Workshop!

Questions & Hints

Summary of Coverage

- Software quality & Software standards
- Reviews/Inspections (static) & Testing (Dynamic)
- Reliability & Security(14) Engineering
- *Software Project Management*

Moderated Exam

Applications Development Theory 4

- **Subject:** APT470S
 - **Date:** Wednesday 05-Nov. @14:00 (FT/PT)
 - **Venue:** M.P.H (Written)
-
- **Sick Test:** 19 Nov 2025 @ 09:00



Review Techniques



What Are Reviews?

- A meeting conducted by **technical people for technical people**
- A **technical assessment** of a work **product** created during the software engineering process
- A software **quality assurance mechanism**



What Reviews Are Not?

- A project summary or progress assessment
- A meeting intended solely to impart information
- A mechanism for political or personal reprisal!



What Do We Look For?

- Errors and defects
 - *Error*—a quality problem found *before* the software is released to end users
 - *Defect*—a quality problem found only *after* the software has been released to end-users



Class Discussion

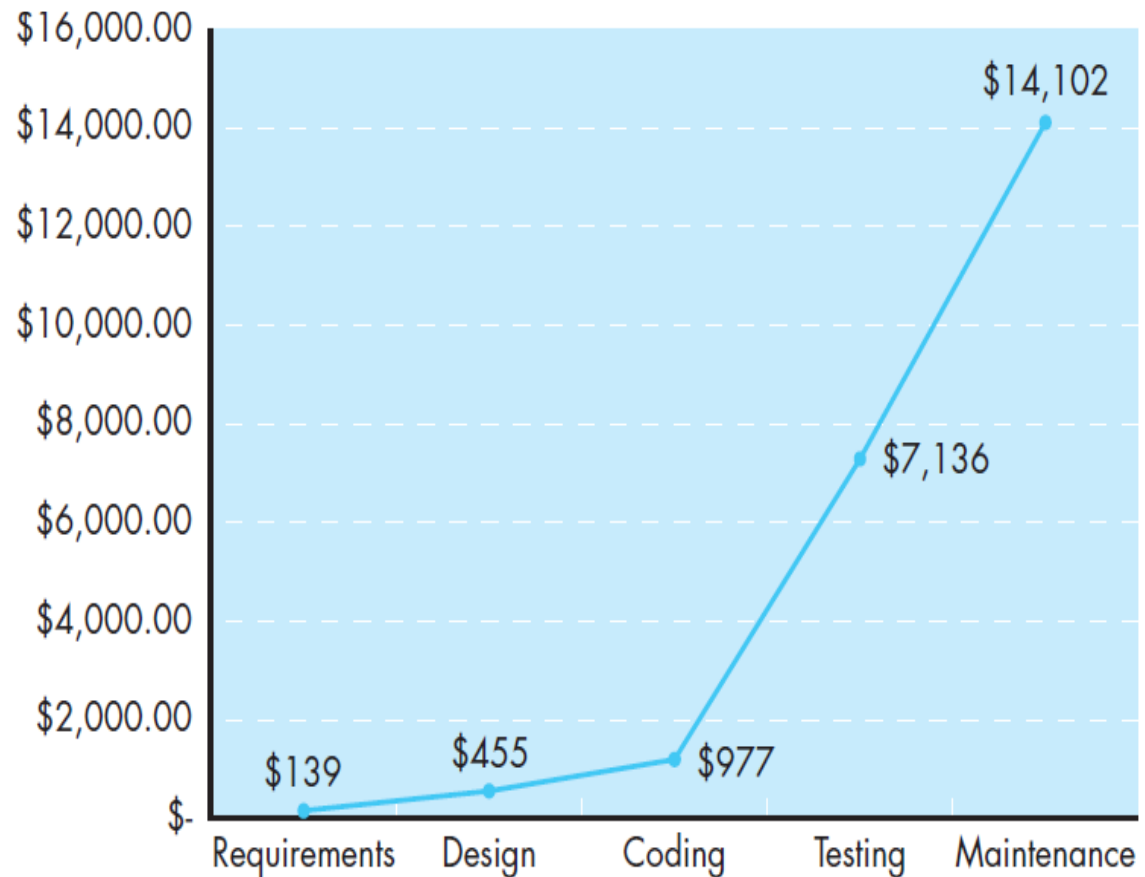
■ *Error*—vs.—*Defect*

- *We make this distinction because errors and defects have very different economic, business, psychological, and human impact*

Relative cost of correcting errors and defects!

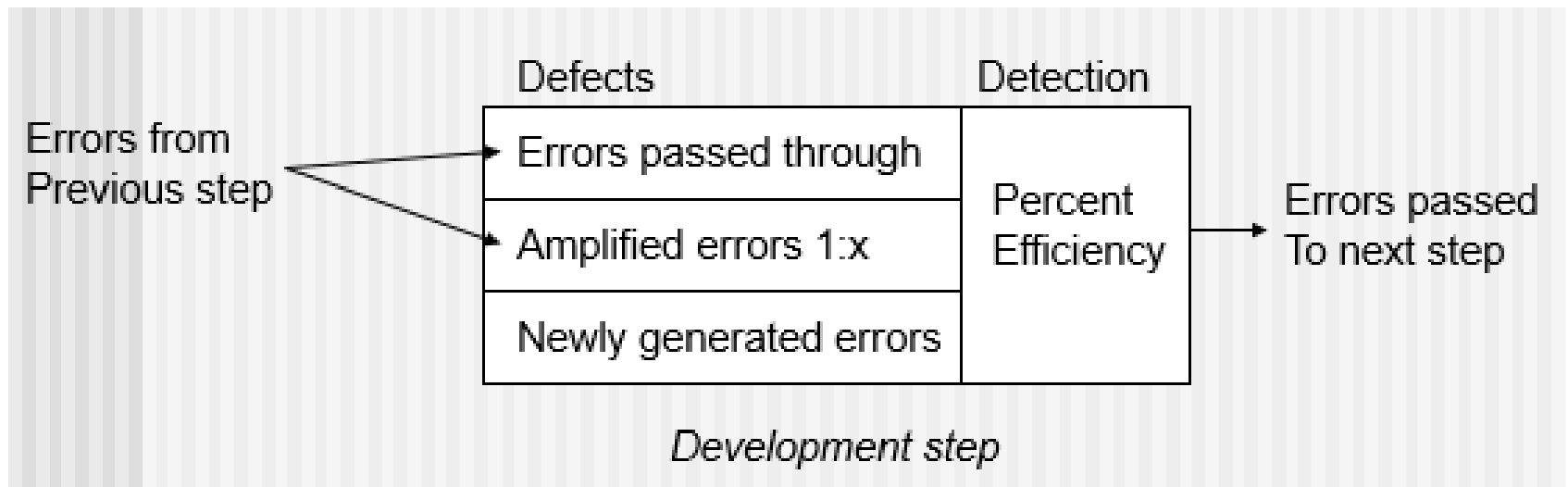
Relative cost
of correcting
errors and
defects

Source: Adapted
from [Boe01b].



Defect Amplification

- **A defect amplification model** [IBM81] can be used to illustrate the **generation** and **detection** of errors during the design and code generation actions of a software process.



Software Reliability Calculation

Scenario:

Assume that 10 errors have been introduced in the requirements model and that each error will be amplified by a factor of 2:1 into design and an addition 20 design errors are introduced and then amplified 1.5:1 into code where an additional 30 errors are introduced. Assume further that all unit testing will find 30 percent of all errors, integration will find 30 percent of the remaining errors, and validation tests will find 50 percent of the remaining errors. (No reviews are conducted).

Reconsider the situation described in the above scenario, but now assume that requirements, design, and code reviews are conducted and are 60 percent effective in uncovering all errors at that step.

Software Reliability Calculation

Solutions:

Calculate Total Errors After Amplification:

From Requirements to Design:

Amplified errors: $10 \text{ req. errors} * 2 = 20 \text{ errors}$

Plus, new design errors: +20

Total Design Errors: $20 + 20 = 40 \text{ errors}$

From Design to Code:

Amplified errors: $40 \text{ design errors} * 1.5 = 60 \text{ errors}$

Plus, new code errors: +30

Total Code Errors (before testing): $60 + 30 = 90 \text{ errors}$

Software Reliability Calculation

Testing:

Unit testing finds 30% of errors: $0.3 \times 90 = 27$ **errors** found. Remaining errors: $90 - 27 = 63$ **errors**.

Integration testing finds 30% of remaining errors: $0.3 \times 63 = 18.9$ **errors**.

Remaining errors: $63 - 18.9 = 44.1$.

Validation testing finds 50% of remaining errors: $0.5 \times 44.1 = 22.05$ **errors**.

Remaining errors: $44.1 - 22.05 = 22.05$.

Errors remaining after all testing: 22.05

— Tutorials —

LTD

Error in the process Software Reliability— **Tutorial**

- Same question but assume that requirements, design, and code reviews are conducted and are 34 percent effective in uncovering all errors at that step. How many errors will be released to the field?

Error in the process Software Reliability— **Tutorial**

- Suppose a software project initially contains 15 errors in its requirements model. Each error will propagate with a factor of 3:1 into the design phase, resulting in an additional 45 design errors. These design errors will then be magnified by a factor of 1.8:1 into the code, introducing an extra 81 errors. Assume that during unit testing, 40 percent of all errors are detected, followed by integration testing uncovering 25 percent of the remaining errors, and finally, validation tests finding 60 percent of the remaining errors. If no reviews are conducted, how many errors will persist and be released into the field?

Reliability Prediction Using Exponential Model — **Tutorial**

Scenario:

A medical monitoring device has a failure rate (λ) of 0.0002 failures/hour.

(a) Calculate the probability that the device operates without failure for 1000 hours.

(c) If regulatory standards require 85% reliability at 1000 hours, does the device pass?

Further Reading!

Prescribed Textbooks:

- (a) Software Engineering, by Ian Sommerville 10th edition, Pearson Education Limited, 2016.
- (b) Software Engineering – A Practitioner's Approach 8th edition Roger Pressman McGraw Hill Education (2015).